AANGARA — Complete Project Analysis

1. Executive Summary & Core Idea

AANGARA is a high-performance, decision-intelligence web application tailored for India's Carbon Credit Trading Scheme (CCTS). The platform transitions deterministic GHG emission intensity accounting and statutory compliance into a strategic capital advantage for heavy industrial entities (cement, steel, aluminum, etc.).

The core idea is to move beyond simple carbon accounting and provide **industrial intelligence**. By utilizing a "Decision Twin," users can benchmark their facilities against empirical peer distributions, visualize their Marginal Abatement Cost (MAC) curves, model future carbon pricing scenarios, and make optimal BUY / BUILD / HYBRID capital allocation decisions to navigate regulatory obligations.

2. Technical Stack

AANGARA is built using a modern, scalable, and highly performant JavaScript/TypeScript ecosystem.

- **Framework:** Next.js 16.3.2 (App Router) powered by Turbopack.
- **Language:** TypeScript (v5.7) for strict type safety.
- **UI Library:** React 19.
- **Styling:** Tailwind CSS (v3.4) with custom PostCSS configurations, utilizing utility classes, CSS variables, and complex glassmorphism effects (`.team-blue-glass-card`, `.glass-shine-card`).
- **State Management:** Zustand (v5.0) for lightweight, predictable global state.
- **Data Visualization:** Recharts (v3.10) for highly interactive, responsive charts (MAC curves, peer benchmarking scatter plots, etc.).
- **Animation Engine:** Native HTML5 Canvas rendering preloaded WebP frame sequences (for the 60fps Fiery Eye hero animation) to ensure zero React re-render thrashing and low CPU/GPU overhead. `OGL` is also present for 3D/WebGL capabilities.
- **Icons:** Lucide React for clean, consistent SVG iconography.
- **PDF Generation:** `jspdf` (v4.2) for generating Boardroom dossiers and ACVA compliance reports directly on the client.
- **Testing:** Vitest (unit tests) and Playwright (E2E testing).

3. Project Architecture & Directory Structure

The codebase follows a feature-driven, modular Next.js App Router pattern:

- **`/app` (Routing & Pages):**
  - `/` (Home): The landing page featuring the hero animation, value propositions, and entry points.
  - `/about`: The core engineering team profiles and mission statement.
  - `/industrial-intelligence`: The primary module for facility input, baseline modeling, and MAC curve generation.
  - `/decision`: The "Decision Twin" cockpit for scenario planning.
  - `/api`: Serverless route handlers for calculating carbon positions (`/api/calculate/carbon-position`), analyzing decarbonization strategies (`/api/intelligence/analyze`), fetching peer benchmarking defaults (`/api/intelligence/defaults/[sector]`), and generating AI explainability (`/api/ai/explain`).
- **`/components` (UI & Features):**
  - `/hero`: Contains `AangaraHeroCore.tsx` — the highly optimized, WebP canvas-driven animation that plays a reveal sequence and transitions into an infinite, organic 14fps living flame loop without clipping or layout shifts.
  - `/intelligence`: Core analysis tools including `FacilityInputForm.tsx` for data ingestion, `DecarbonisationMatrix.tsx` (rendering the MAC Curve), and `PeerBenchmarkCard.tsx`.
  - `/cockpit`: "Decision Twin" interactive widgets including `ScenarioSliders.tsx` (carbon price forecasting), `CarbonPositionCard.tsx` (financial exposure), and `ExplainabilityCard.tsx`.
  - `/about`: `TeamSection.tsx` which renders the compact portrait team cards with blue-tinted glassmorphism, fully visible initials avatars, and direct brand icons.
  - `/ui` & `/charts`: Reusable atomic design components and Recharts wrappers.
- **`/lib` (Core Logic):**
  - Contains formatters (INR Cr, kt CO_2e, payback years), NPV/IRR financial calculators, and `exportPdf.ts` for handling client-side PDF document generation.
- **`/public` (Static Assets):**
  - Contains the high-resolution transparent AANGARA logo and the `/images/hero_frames/` directory containing the 50 sequential WebP frames for the hero canvas animation.

4. Key Features & Implementation Details

A. The Decision Twin Cockpit

- **Implementation:** Found in `components/cockpit`. It acts as the financial command center.
- **Details:** Uses interactive sliders (`ScenarioSliders.tsx`) to adjust parameters like future CCC market prices (₹/t) and production capacity. It recalculates the net carbon position (Short/Long) dynamically and translates it into direct financial exposure (INR Crores).

B. Marginal Abatement Cost (MAC) Curve

- **Implementation:** `DecarbonisationMatrix.tsx` using `recharts`.
- **Details:** Evaluates potential engineering projects (e.g., WHRS, LC3 Transition, Solar-Wind). Projects are sorted left-to-right by marginal abatement cost (₹/tCO_2e) and plotted as a bar chart. Bars are color-coded by payback tier (Fast < 2.5y, Medium, Strategic > 4.5y). Includes a reference line for the CCC market baseline. A top summary strip calculates total portfolio abatement potential, required CAPEX, and 10-Yr NPV.

C. Industrial Benchmarking

- **Implementation:** `FacilityInputForm.tsx` & `PeerBenchmarkCard.tsx`.
- **Details:** Users input facility parameters (e.g., Clinker factor, specific heat consumption). The system fetches empirical sector distributions via `/api/intelligence/defaults` and plots the facility's emission intensity against the industry median and top 10% percentiles, generating a compliance baseline gap.

D. High-Performance Visuals

- **Implementation:** `AangaraHeroCore.tsx` & `globals.css`.
- **Details:** The hero section eschews heavy WebGL or unoptimized videos for a smart Canvas + WebP sequence. All 50 frames are preloaded. A `requestAnimationFrame` loop progresses via timestamp and cubic ease-out, settling into a continuous oscillation (frames 16-38) to simulate a living flame. Visibility guards (`IntersectionObserver`, `visibilitychange`) pause the RAF loop when off-screen to ensure zero CPU/GPU drain. The team cards use `.team-blue-glass-card` CSS with `-webkit-backdrop-filter` and advanced pseudo-element shine animations.

E. Export & Reporting

- **Implementation:** `lib/exportPdf.ts` utilizing `jspdf`.
- **Details:** Converts the interactive web dashboards into static, formatted ACVA-compliant dossiers and Boardroom presentations. Recently scrubbed of all legacy placeholder branding across headers, footers, and metadata to strictly reflect the AANGARA brand.

5. Conclusion

AANGARA is a meticulously engineered, specialized SaaS application. It successfully bridges complex environmental compliance (India's CCTS) with corporate finance (CAPEX, NPV, MAC). The architecture is modern, strictly typed, visually polished, and highly optimized for both performance and complex data visualization.